Methodology: The CIF-AD-OODA Integration Model

Methodology: The CIF-AD-OODA Integration Model I

This section establishes the analytical framework used to evaluate cognitive integrity threats across the ten critical domains examined in this paper. We integrate three complementary theoretical frameworks—the Cognitive Integrity Framework (CIF) [**?**], Axiomatic Design (AD) [**?**, **?**], and the OODA Loop [**?**]—into a unified model for analyzing Goal Hijacking attacks and their defenses.

Framework Integration I

Each framework contributes a distinct analytical dimension:

The Cognitive Integrity Framework (CIF) provides the defense mechanism vocabulary. Developed formally in Paper 1 of this series [**?**] and validated computationally in Paper 2 [**?**], CIF defines five canonical defense mechanisms that protect agent cognitive states $\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{I}_i, \mathcal{H}_i \rangle$ against adversarial manipulation. These mechanisms operate at the architectural, runtime, and coordination layers to maintain belief consistency, goal alignment, and provenance verifiability.

Axiomatic Design (AD) provides the structural model. Suh's Independence Axiom [**?**] states that a good design maintains the independence of Functional Requirements (FRs): solving for FR_1 must not compromise FR_2. We represent this as the Design Matrix equation:

Framework Integration II

where $\{FR\}$ is the vector of Functional Requirements, $\{DP\}$ is the vector of Design Parameters, and $[A]$ is the Design Matrix. In an uncoupled (safe) design, $[A]$ is diagonal. Goal Hijacking attacks introduce off-diagonal terms, coupling previously independent requirements.

$$\{FR\} = [A]\{DP\} \tag{1}$$

The OODA Loop provides the temporal model. Boyd's Observe-Orient-Decide-Act cycle [**?**, **?**] captures the decision-making process of autonomous agents. Goal Hijacking targets the **Orient** phase—the synthesis stage where agents integrate observations with prior knowledge, training data, and system prompts. By corrupting Orientation, adversaries redirect the downstream Decide and Act phases while preserving the illusion of rational behavior.

Framework Integration III

The integration insight is that Goal Hijacking operates simultaneously across all three dimensions: it introduces transient coupling in the Design Matrix (AD), corrupts the Orient phase (OODA), and is detectable and defensible through the canonical mechanisms (CIF).

CIF Defense Mechanisms I

Paper 1 [**?**] defines five canonical defense mechanisms. We reproduce them here for reader convenience, as they form the defense vocabulary applied across all ten domains:

Mechanism	Formal Notation	Function
Cognitive Firewall	$\mathcal{F}(m) \to \{\text{accept}, \text{quarantine}, \text{reject}\}$	Classifies incoming messages by trust score against threshold τ; quarantines ambiguous inputs for sandboxed evaluation

CIF Defense Mechanisms II

Belief Sandboxing	$\mathcal{B}_{\text{verified}}/\mathcal{B}_{\text{provisional}}$ partition	Isolates unverified beliefs from the agent's operational belief set; requires corroboration for promotion
Behavioral Invariants	Predicates $\text{INV}_k(\sigma_i) \in \{\text{true}, \text{false}\}$	Pre-defined runtime checks (e.g., goal alignment, permission boundaries) that trigger alerts on violation
Drift Detection	$S_{\text{drift}} = (\mathcal{B}_i^t \| \mathcal{B}_i^{t-1})$	Monitors belief distribution changes via KL

CIF Defense Mechanisms III

Byzantine Consensus	$\mathcal{B}_{\text{consensus}}$ with quorum q, requiring $n \geq 3f+1$	Multi-agent agreement protocol tolerating up to f compromised agents among n total

CIF Defense Mechanisms IV

These mechanisms compose in series and parallel to achieve layered defense. Paper 2 [**?**] demonstrates that the recommended defense stack achieves 94–100% detection at the parametric design ceiling across 950 attack scenarios and four production multiagent architectures. Paper 3 [**?**] translates these results into deployment guidance, monitoring playbooks, and cost–benefit frameworks; readers seeking engineering guidance on instantiating the mechanisms below in production should consult Part 3.